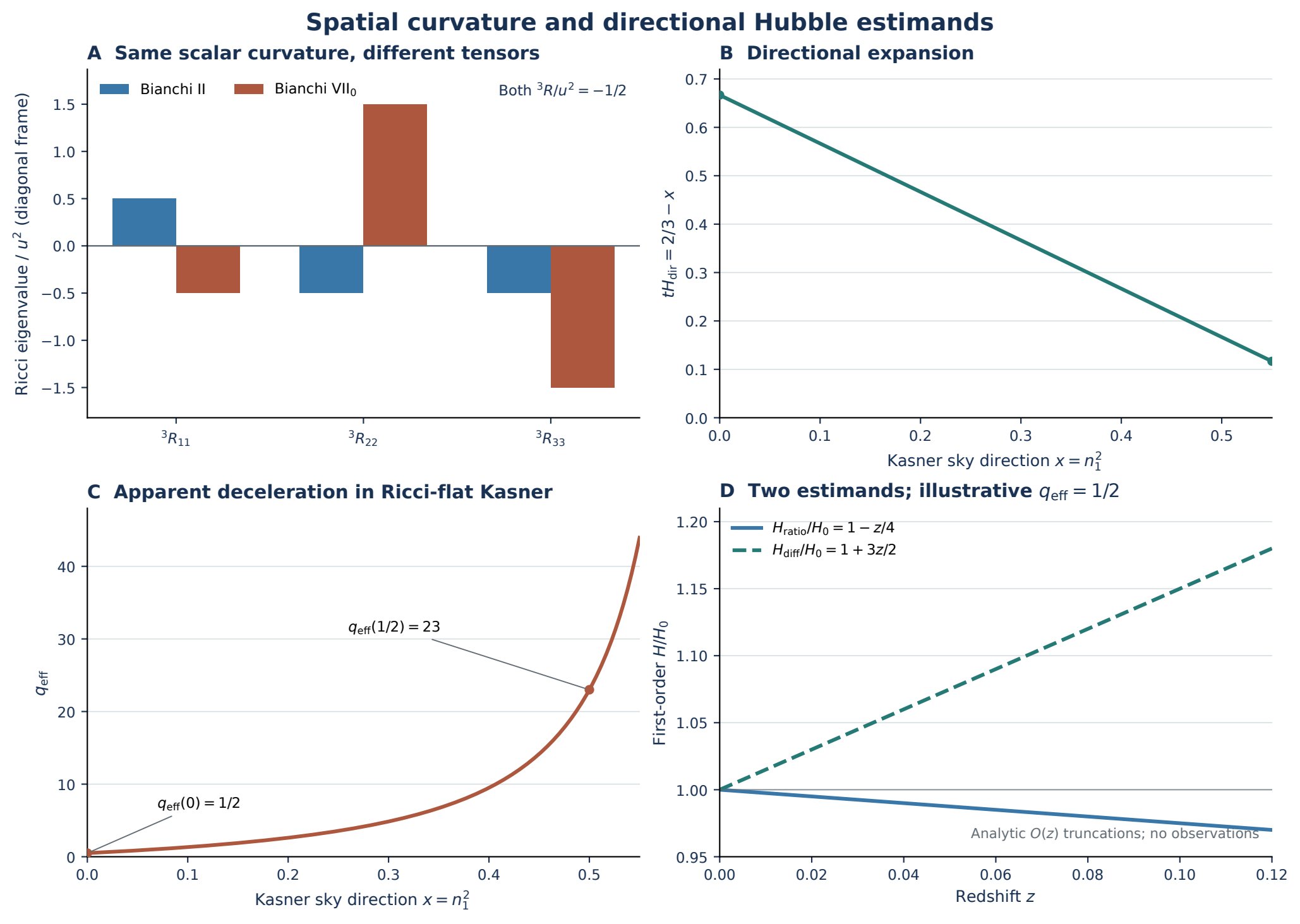
Spatial curvature and directional Hubble estimands
A Same scalar curvature, different tensors
Bianchi II
Bianchi VII$_0$
Both ${}^3R/u^2 = -1/2$
Ricci eigenvalue / u^2 (diagonal frame)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
${}^3R_{11}$
${}^3R_{22}$
${}^3R_{33}$
B Directional expansion
$tH_{\rm dir} = 2/3 - x$
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.0
0.1
0.2
0.3
0.4
0.5
Kasner sky direction $x = n_1^2$
C Apparent deceleration in Ricci-flat Kasner
$q_{\rm eff}$
40
30
20
10
0
$q_{\rm eff}(1/2) = 23$
$q_{\rm eff}(0) = 1/2$
0.0
0.1
0.2
0.3
0.4
0.5
Kasner sky direction $x = n_1^2$
D Two estimands; illustrative $q_{\rm eff} = 1/2$
$H_{\rm ratio}/H_0 = 1 - z/4$
$H_{\rm diff}/H_0 = 1 + 3z/2$
First-order H/H_0
1.20
1.15
1.10
1.05
1.00
0.95
Analytic $O(z)$ truncations; no observations
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Redshift z